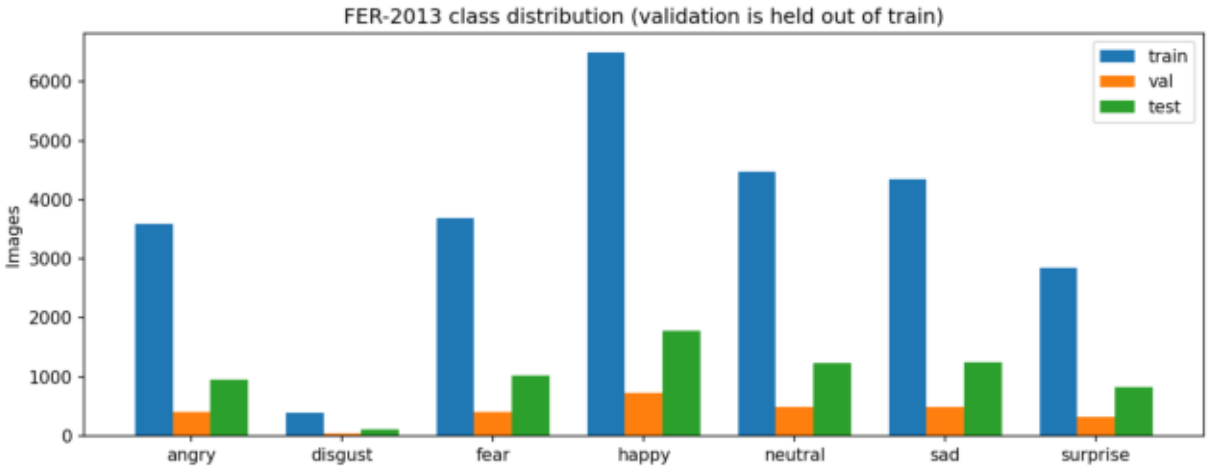
Facial emotion recognition - methods

Data: msambare/fer2013 on Kaggle. Seven emotions; grayscale 48 x 48 pixels. No supplementary training
dataset is used. ResNet-18 starts from ImageNet-1K weights.

Split protocol: validation is stratified from the original training folder; the supplied test folder is reserved for final evaluation. The saved split manifests record every file and label. Model selection uses validation accuracy, with macro F1 as a tie-breaker.

Split sizes: train=25,838, val=2,871, test=7,178.
Class order: angry, disgust, fear, happy, neutral, sad, surprise.



Preprocessing: convert to grayscale, resize to 48 x 48, scale pixels to [0,1], then normalize with mean=0.5 and std=0.5. Training adds horizontal flips and mild affine rotation, shift and scale. Validation, test and realtime inference use no augmentation. Square-root inverse class-frequency loss weights are derived only from training labels. Lighting profiles add brightness/contrast jitter; occlusion adds small random erasing. Gentle combined uses reflect-padded crops, smaller affine changes
and occasional mild jitter. Selected profiles: M1=gentle_combined, M2=occlusion.

M1 (from scratch): four Conv3x3-BatchNorm-ReLU-MaxPool2x2-Dropout blocks, with 32/64/128/256 channels
and dropout 0.10/0.15/0.20/0.25. Flatten 3x3x256=2304, Linear256-BatchNorm-ReLU-Dropout0.50, then
Linear7.

M2 (transfer): ImageNet ResNet-18; sum the first convolution's RGB weights to accept one channel, set its stride to 1 and remove the initial max pool to preserve small-image detail. Replace the head with Dropout0.30-Linear7 and fine-tune every layer. The input normalization is deliberately adapted for FER, rather than the stock ImageNet transform.

Both models return logits for cross entropy; softmax is applied at inference.

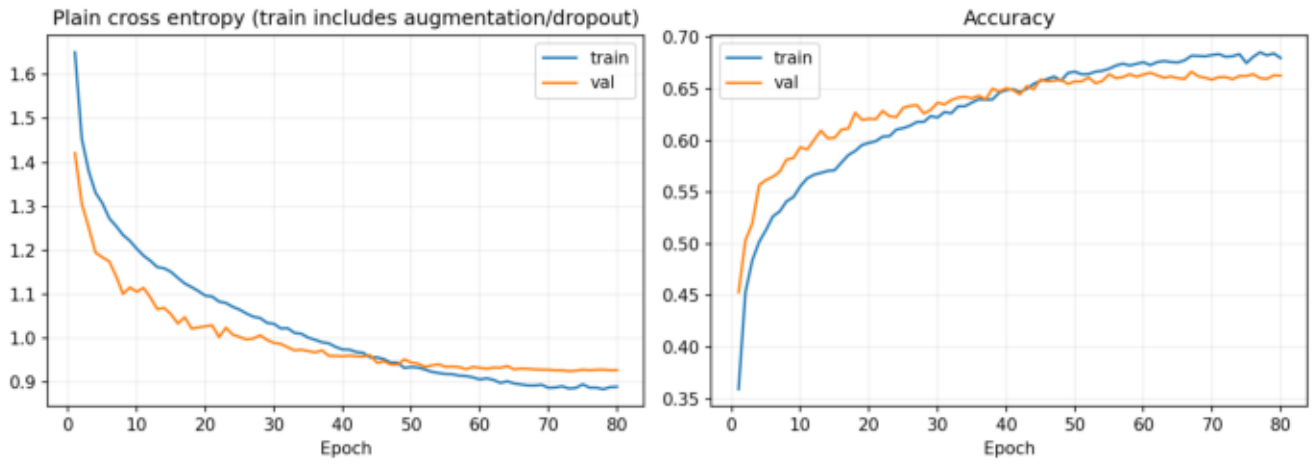
Training and quantitative comparison

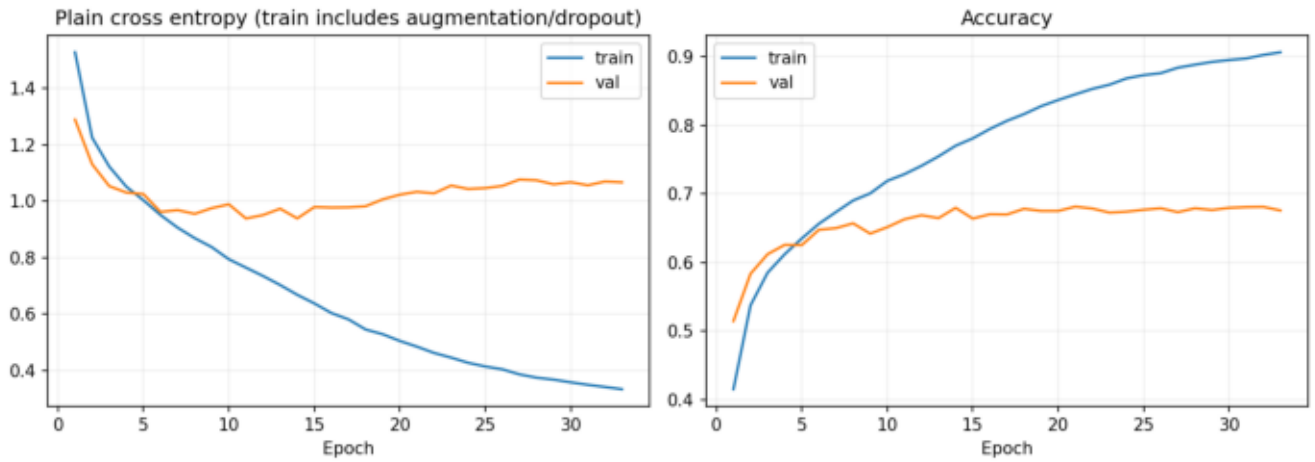
CNN: AdamW lr=0.001, backbone lr=n/a, weight decay=0.0001, smoothing=0.05, epochs <= 80, patience=18,
seed=42. Cosine LR schedule. Best epoch=67. Training device=cuda. Augmentation=gentle_combined.

TRANSFER: AdamW lr=0.001, backbone lr=5e-05, weight decay=0.0005, smoothing=0.05, epochs <= 40, patience=12, seed=42. Cosine LR schedule. Best epoch=21. Training device=cuda. Augmentation=occlusion.

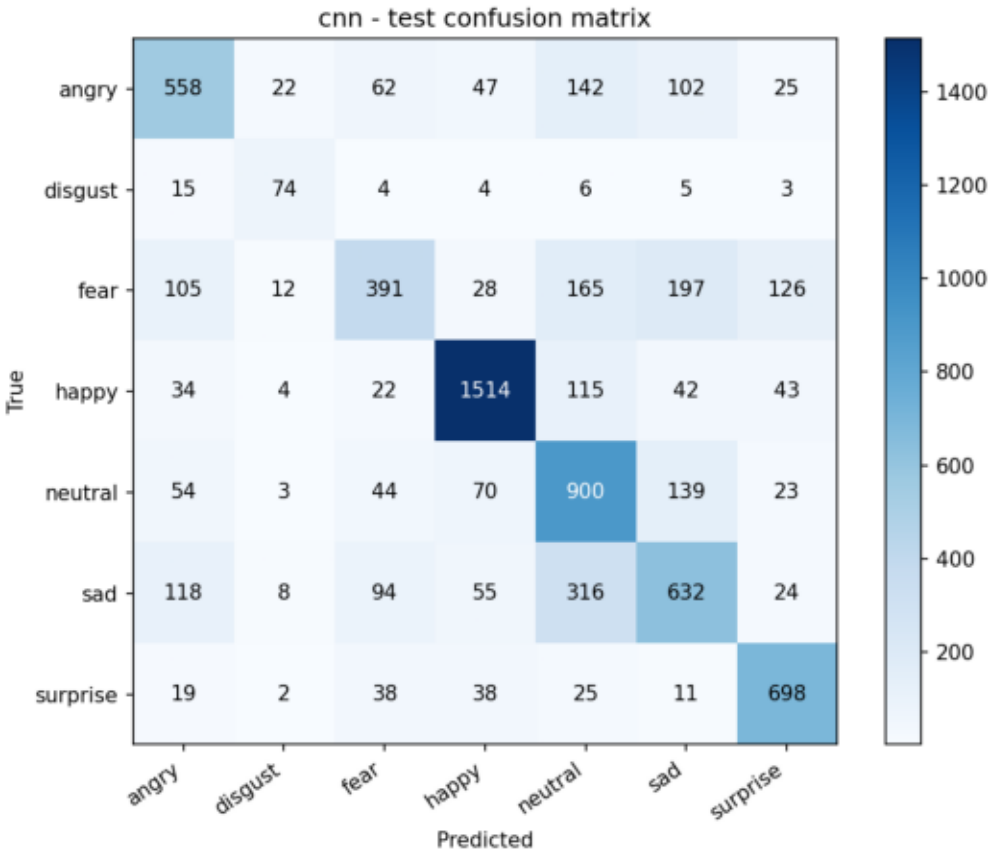
Model	Accuracy	Macro F1	Params	CPU FPS	Target met
cnn	66.41%	0.6415	980,711	486.9	yes
transfer	66.17%	0.6488	11,173,831	49.1	yes

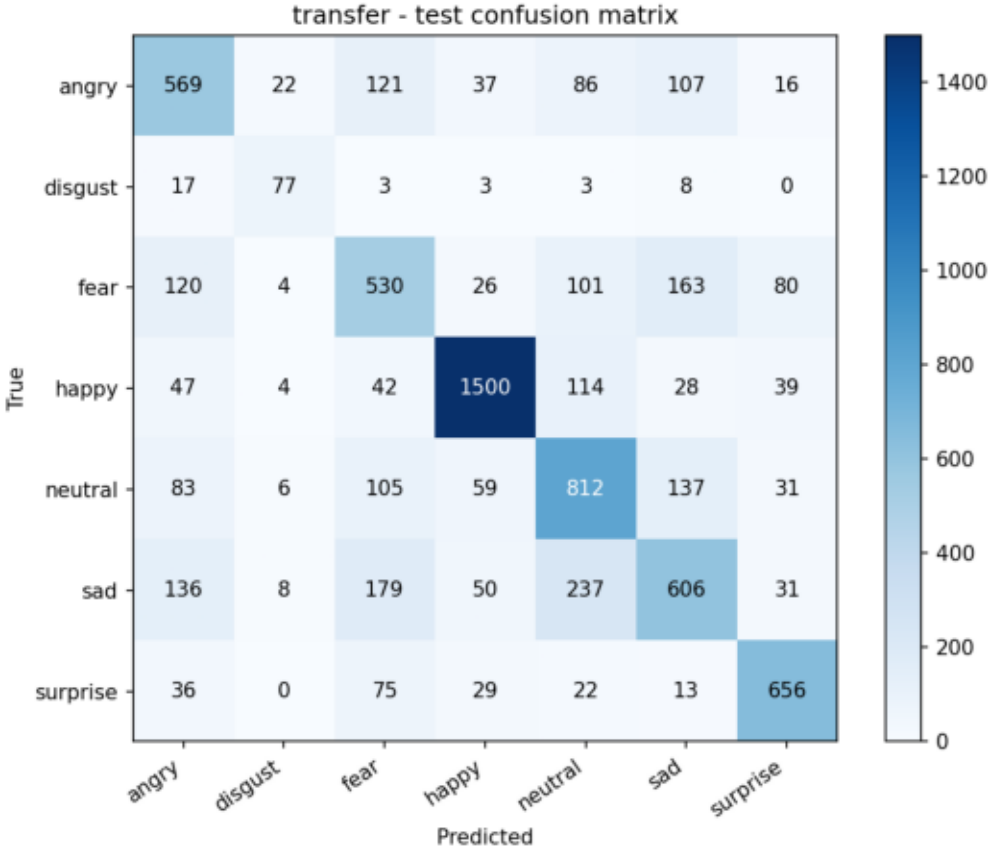
Targets: M1 >= 60%, M2 >= 65%. CPU FPS is batch-one model forward only (float32, 1x48x48); it excludes preprocessing, face detection and camera I/O. Realtime throughput must be measured separately on the demonstration laptop. Benchmark CPU: x86_64, threads=4.





Confusion matrices on held-out test data





CNN - misclassified faces and analysis



1. angry -> sad. Assigned sad 79.1%, versus angry 4.5% (gap 74.6%). Inspect the shared facial cues and whether this label is visually ambiguous.

2. disgust -> surprise. Assigned surprise 61.3%, versus disgust 2.3% (gap 59.0%). Inspect the shared facial cues and whether this label is visually ambiguous.

3. fear -> sad. Assigned sad 60.8%, versus fear 10.1% (gap 50.7%). Inspect the shared facial cues and whether this label is visually ambiguous.

4. happy -> angry. Assigned angry 41.2%, versus happy 15.4% (gap 25.8%). Inspect the shared facial cues and whether this label is visually ambiguous.

5. neutral -> angry. Assigned angry 67.9%, versus neutral 10.5% (gap 57.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

6. sad -> fear. Assigned fear 31.5%, versus sad 22.3% (gap 9.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

7. surprise -> happy. Assigned happy 63.5%, versus surprise 23.7% (gap 39.9%). Inspect the shared facial cues and whether this label is visually ambiguous.

8. angry -> fear. Assigned fear 42.6%, versus angry 33.9% (gap 8.7%). Inspect the shared facial cues and whether this label is visually ambiguous.

9. angry -> neutral. Assigned neutral 33.3%, versus angry 22.2% (gap 11.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

10. angry -> sad. Assigned sad 41.5%, versus angry 26.2% (gap 15.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

Observations above describe score errors and suggest visual checks. Add your own image-specific observations before submission; score patterns alone cannot establish why a face was misclassified.

TRANSFER - misclassified faces and analysis



1. angry -> sad. Assigned sad 98.2%, versus angry 0.7% (gap 97.5%). Confident error; inspect label ambiguity, face crop and domain mismatch.

2. disgust -> fear. Assigned fear 75.1%, versus disgust 1.7% (gap 73.4%). Inspect the shared facial cues and whether this label is visually ambiguous.

3. fear -> angry. Assigned angry 67.5%, versus fear 6.4% (gap 61.1%). Inspect the shared facial cues and whether this label is visually ambiguous.

4. happy -> surprise. Assigned surprise 78.6%, versus happy 16.7% (gap 62.0%). Inspect the shared facial cues and whether this label is visually ambiguous.

5. neutral -> fear. Assigned fear 60.1%, versus neutral 15.5% (gap 44.6%). Inspect the shared facial cues and whether this label is visually ambiguous.

6. sad -> surprise. Assigned surprise 96.9%, versus sad 0.1% (gap 96.8%). Confident error; inspect label ambiguity, face crop and domain mismatch.

7. surprise -> happy. Assigned happy 67.5%, versus surprise 13.5% (gap 54.0%). Inspect the shared facial cues and whether this label is visually ambiguous.

8. angry -> fear. Assigned fear 39.9%, versus angry 24.3% (gap 15.6%). Inspect the shared facial cues and whether this label is visually ambiguous.

9. angry -> fear. Assigned fear 52.7%, versus angry 5.5% (gap 47.2%). Inspect the shared facial cues and whether this label is visually ambiguous.

10. angry -> happy. Assigned happy 41.6%, versus angry 21.1% (gap 20.5%). Inspect the shared facial cues and whether this label is visually ambiguous.

Observations above describe score errors and suggest visual checks. Add your own image-specific observations before submission; score patterns alone cannot establish why a face was misclassified.

Realtime system and demonstration

Pipeline: OpenCV VideoCapture -> Haar frontal-face detection -> crop each face with a small margin ->
shared grayscale preprocessing -> batched PyTorch inference -> softmax -> bounding boxes, labels,
confidence and FPS.

The command line selects cnn/transfer and accepts a phone URL, webcam, image or file video. Connection
failures trigger bounded retries and a clear error. Low-confidence predictions are labelled uncertain.
Press s to save the phone-camera demonstration.

The on-screen FPS includes capture and processing, but excludes display and output encoding. It depends on image resolution, number of faces, CPU threads and Wi-Fi. The 10 FPS CPU requirement must
be checked on the student's own laptop.

DEMONSTRATION PENDING: no real phone-camera screenshot was supplied. Run the app, press s, then
regenerate this report with --screenshot. This report is a draft until the demonstration and visual error analysis are completed.

Limitations: FER labels can be ambiguous; a visible expression does not reliably reveal a person's internal emotional state. Haar detection is strongest on frontal faces. Lighting, pose, occlusion and the FER-to-phone domain shift can reduce accuracy. Softmax scores are model confidence values, not calibrated probabilities.

Sources: kaggle.com/datasets/msambare/fer2013; torchvision ImageNet ResNet-18 (docs.pytorch.org/vision/stable/models/generated/torchvision.models.resnet18.html). Checkpoints, split manifests, class counts, run configuration, CSV metrics and error examples accompany the experiment for reproducibility.